

Proposal for Exposition Paper

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BSM CS3A - Graph Test Auto

Title

Blood Distribution from Blood banks to Medical Offices with the Help of Network Flow Model

Overview of the Topic

Health is invaluable and there are people in bad shape who often need the aid of those fortunate enough who are able to help. There are many things in today's society that can threaten the well being of people and one of the crucial things they potentially need is **blood**. **Blood supply** is able to help in a lot of dire situations, from blood loss due to injuries, all the way to cancer patients recovering from heavy treatments such as chemotherapy. There are always people who need help and oftentimes the **demand is too high** for the supply to keep up. That is why in this study the researchers aim to aid the said problem by helping **blood banks** be able to **distribute sufficient amounts** of blood supply to hospitals and other medical facilities. All while applying methods of **graph theory** to find a systematic way of **distributing** the said supply.

Objectives

The objective of this study is to develop a model that optimizes the distribution of blood to hospitals across Malolos to ensure **efficient**, **equitable**, and **demand-responsive**

delivery. Specifically, this study aims to:

- Represent the blood bags distribution system as a graph, where **blood banks** and **hospitals** are modeled as vertices and **transportation routes** as weighted edges
- **Maximize total satisfied blood demand** across hospitals, subject to blood supply availability and transportation capacity constraints.
- **Minimize transportation costs** while meeting hospital demand requirements
- Evaluate Availability of blood types, **ensure that high demand blood types are prioritized** to minimize shortages during normal operating conditions

Transformation to Graph Theory Problem

The distribution of blood supplies among the hospitals and the blood centers is represented into a graph theory problem to analyze and reduce blood shortages. The blood distribution system is modeled as a **directed graph** $G = (V, E)$, where each **node** represents a hospital or blood centers, and each **directed edge** represents an allowable route between facilities.

The graph is **directed** to represent the constraints in healthcare logistics, such as predefined delivery routes, policies, and non-reciprocal transportation agreements, rather than physical road directionality.

The **Nodes** in the graph are defined according to their function within the blood distribution network to represent the **Supply-Demand Structure**

- **Supply Nodes** represents blood centers that provide blood units.
- **Demand Nodes** represents hospitals or facilities that require blood units
- **Transshipment Nodes** which may both receive and redistribute blood units.

Each node maintains an inventory of the eight blood types: A+, A-, B+, B-, AB+, AB-, O-, O+.

Each weight of a **directed edge** is normalized using **distance across the nodes** as primary measure to ensure consistency across the network and avoid errors between distance-based and time-based interpretations.

The graph theory problem is addressed in this study is a **network flow** and system balance problem, where the objective is to determine the efficient distribution path and transfer quantities that **minimize blood shortages** and to reduce preventable deaths across all nodes in the network.

Methodology

The scope of this study is limited only to **Malolos City, Bulacan**, where blood banks and hospitals are densely distributed and emergency demand is significant for emergency blood distribution. The node scope consists of all identified **blood banks, hospitals, and clinics** operating within Malolos City at the time of data collection. Each facility is represented as a vertex in the graph. Blood banks are classified as supply nodes, while hospitals and clinics are classified as demand nodes.

The edge scope includes **road-based transportation** routes connecting the identified facilities. Each edge represents a traversable road segment between two facilities and is directed to reflect feasible travel directions. Only primary and secondary roads commonly used for medical logistics are considered, excluding pedestrian-only paths and non-road transport modes. The data used for finding the distance and road connectivity will be found by using **OpenStreetMap**, while travel-time estimates and route validation are cross-checked using **Google Maps**.

The data sources for this study include local health office records for facility identification, OpenStreetMap for road network topology, and Google Maps for route verification and travel-time approximation. The model uses a **static time snapshot**, representing typical weekday operations under normal traffic and demand conditions. Temporal variations such as hourly traffic fluctuations and emergency surge dynamics are not incorporated in this phase of the study.

Here is the **weighted graph** made from the hospitals and blood banks as vertices and the transportation route as the weighted and directed edges in Malolos, Bulacan.

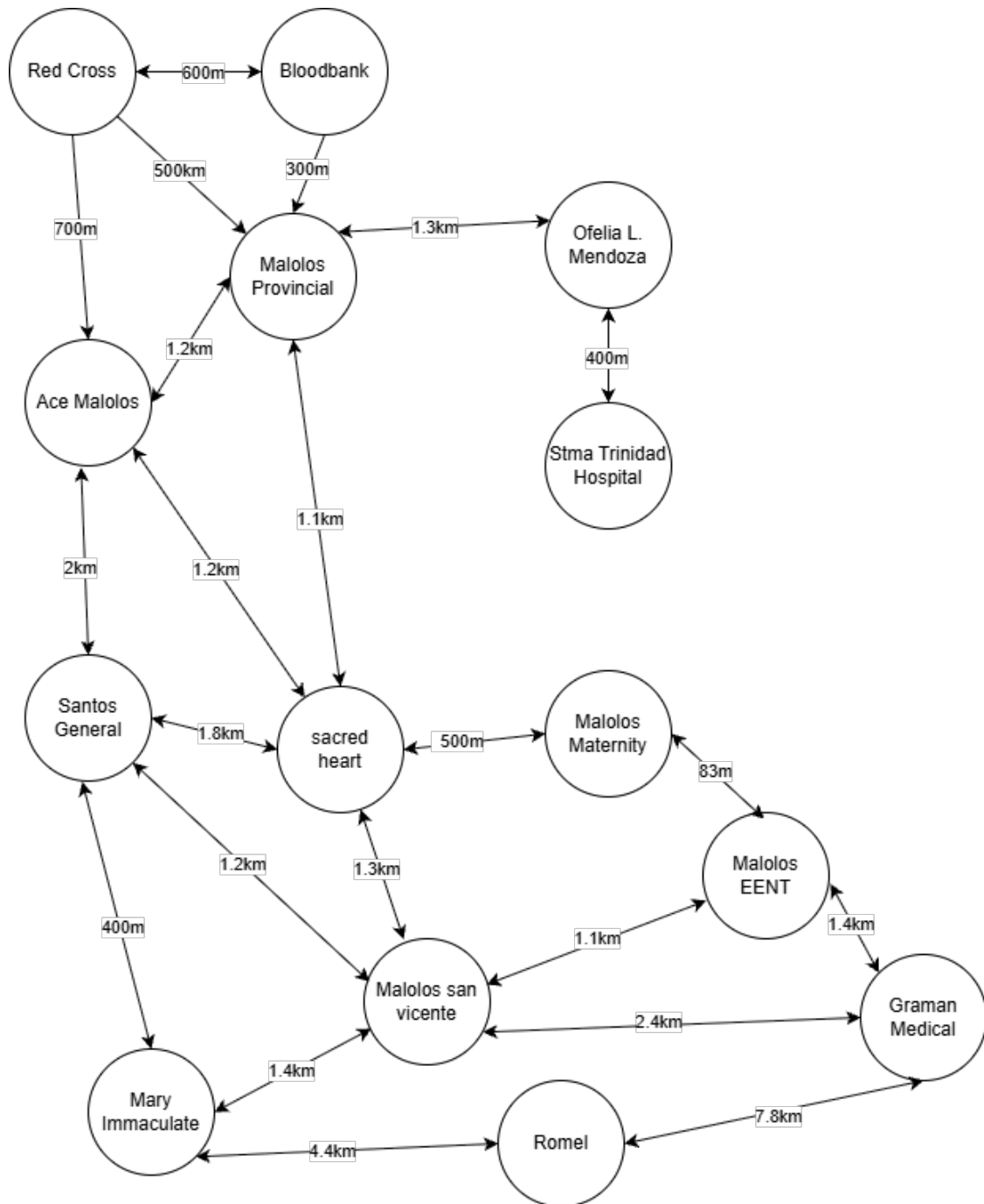


Figure 1: Connection Graph

Expected Outcomes

The researchers expect to create a **graph** based on the blood bag **distribution network** in Malolos to visualize the flow of blood supplies. The study aims to develop an **optimal distribution plan** that ensures different blood types to be distributed evenly and efficiently between hospitals based on their needs. The expected results of this study include improvements in **distribution efficiency**, like a higher percentage of hospital demand being met and also the identification of important facilities and distribution links that would affect the flow of blood supplies. In addition to this, the result of this study may become a helpful reference for future research related to blood distribution.

References

1. Altunoglu, B., & Batur Sir, G. D. (2024). *Multi-objective location-distribution optimization in blood supply chain: an application in Turkiye*. BMC Public Health. <https://paperity.org/p/359956709/multi-objective-location-distribution-optimization-in-blood-supply-chain-an-application>
2. Ala, A., Simic, V., Nebojsa Bacanin, & Erfan Babae Tirkolae. (2024). *Blood supply chain network design with lateral freight: A robust possibilistic optimization model*. Engineering Applications of Artificial Intelligence, 133, 108053–108053. <https://doi.org/10.1016/j.engappai.2024.108053>
3. Ahuja, R. K., Magnanti, T. L., & Orlin, J. B. (1993). *Network flows: Theory, algorithms, and applications*. Prentice Hall. <https://www.pearson.com/en-us/subject-catalog/p/network-flows-theory-algorithms-and-applications/P200000003456/9780136175490>
4. Morella, J. M., & Melvin, T. R. (2024). *Graph-based modeling of emergency logistics networks under urban constraints*. Journal of Transportation and Health, 32, 101–113

[https://www.staff.ces.funai.edu.ng/papersCollection/scholarship/_pdfs/
Network_Flows_Theory_Algorithms_And_Applications_Ravindra_K_Ahuja](https://www.staff.ces.funai.edu.ng/papersCollection/scholarship/_pdfs/Network_Flows_Theory_Algorithms_And_Applications_Ravindra_K_Ahuja)